SUMMARY OF MASTER'S RESEARCH

Title: Energy Performance Optimization of Integrated Passive Envelope for Mid-Rise Office Buildings in Tropical Climate

Objective and Problem Statement: Passive envelope strategies differ markedly in their maturity for tropical offices. External louvers and, to a growing extent, double-skin facades (DSF) have been studied and applied in hot-humid office contexts, whereas windcatchers remain largely at the research stage, assessed in residential or general building types rather than offices and concentrated outside the tropics. Across all three, the existing research is dominated by numerical simulation (reported in the thesis as 79.3% of reviewed studies, against 6.9% with experimental validation) and evaluates each technology in isolation: windcatcher studies focus almost exclusively on aerodynamic air-change performance, louver studies on daylighting and sensible heat, and only one of the reviewed studies addresses the relative-humidity and latent-load penalties that are decisive in hot-humid climates. Holistic assessments of the three combined as an integrated passive envelope remain scarce, particularly for Jakarta. This study addressed that gap, asking: to what thermodynamic extent can an integrated passive envelope, a deep-cavity DSF, an aerodynamic windcatcher empirically recharacterized as a thermal stack tower (TST), and an optimized louver system, mitigate the sensible cooling load in Jakarta's equatorial climate before reaching its aerodynamic and hygric performance boundaries? The aim was to define the precise threshold at which purely passive operation becomes insufficient and active mechanical supplementation is required.

Methodology: A two-phase parametric optimization was developed in DesignBuilder and EnergyPlus, fully coupled with an airflow network (AFN) diagnostic, and applied to a six-storey, 2,400 m^2 Jakarta office archetype (ASHRAE 90.1 prototype; 40% window-to-wall ratio; south-facing) over a full annual cycle (8,760 hours) using the Jakarta-Kemayoran EPW dataset. Phase 1 assessed 32 component-level cases (10 DSF, 10 windcatcher, 12 louvers) and ranked the governing geometric variables through a range (*R*) sensitivity method, with post-processing conducted in Python. Phase 2 synthesized the component optima into three integrated scenarios – optimal, moderate, and worst-case – selected through multi-criteria decision analysis (MCDA). The AFN coupling was the decisive methodological choice, as conventional thermal simulation assumes fixed air-change rates and conceals the interaction between buoyancy-driven cavity flow and the fresh-air supply required by occupancy. The EnergyPlus-AFN approach for multi-storey naturally ventilated DSF systems has

been experimentally validated in prior work (Andelković et al., 2016); the present study, however, was simulation-based and did not include experimental validation of its own model – a boundary condition acknowledged in the thesis.

Key findings: the optimal configuration (scenario 1: a 1.5 m cavity, a 9 m long-rectangular TST, and permeable louver array) performed effectively as a sensible thermal buffer, achieving a 30.3% reduction in annual sensible cooling demand (cooling EUI reduced from 146.51 to 102.14 kWh/m^2/year) and a 26.47% reduction in total EUI (from 149.80 to 110.14 kWh/m^2/year) once the artificial-lighting penalty was included. The AFN diagnostic, however, exposed a failure that the energy results alone would conceal. The heated DSF cavity operated in dominant exhaust mode (-1.50 m^3/s), acting as a parasitic vacuum that recirculated internal air at 0.392 ach while the genuine outdoor air captured by the TST stalled at 0.073 ach – an 89.3% deficit against the ASHRAE 62.1 minimum of 0.68 ach, with the cavity exhausting 5.41 times more air than the TST supplied. This behavior also sustained an unresolved latent dehumidification load (3,553.99 W; mean indoor relative humidity 55.86%). A quadrant analysis confirmed the deadlock quantitatively: each percentage point of comfort gain was accompanied by a disproportionate dual stress (high-humidity) penalty, in a fixed 1.85:1 ratio. The central conclusion is that, in hot-humid climates, optimizing a passive envelope for sensible efficiency is inversely correlated with fresh-air delivery and latent-load containment; passive thermal optimization and ventilation sufficiency are thermodynamically irreconcilable within a purely passive framework.

Contribution: The work introduced two diagnostic frameworks. The first is the aerodynamic short-circuit ratio (ASR), an AFN-derived metric defined as the proportion of total internal air circulation attributable to recirculated rather than genuine outdoor air (81.4% in the optimal scenario), providing a precise instrument for diagnosing fresh-air-delivery failure in integrated passive envelopes. The second is a three-pathway heat-gain decomposition that disaggregates the sensible cooling load into solar-radiative, conductive, and internal constituents, validated by arithmetic closure against EnergyPlus zone-level outputs. On this empirical basis, the study proposed the thermal-hygric adaptive decoupled envelope system (THADES): a framework that designates the DSF-louver skin strictly as a solar interceptor and sensible heat buffer, and delegates fresh-air flushing and latent-load management to targeted low-energy mechanical supplementation, such as a dedicated outdoor-air system or low-energy exhaust integrated within the TST shaft. The principal personal contributions were the development and calibration of the full EnergyPlus-AFN

methodology, the diagnosis of the short-circuit mechanism, and the formulation of the ASR, the heat-gain decomposition, and the THADES construct. This diagnosis is precisely where my proposed doctoral work begins: resolving, at the flow-field level, why the short-circuit occurs and how the envelope can be re-engineered to break it, which the attached research proposal sets out in detail.